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United States Patent	12408958
Kind Code	B2
Date of Patent	September 09, 2025
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Distal radius stabilization system

Abstract

Devices, systems, and methods for bone stabilization, especially ulna head stabilization. The stabilization system may include a bone plate having an elongated portion extending along a longitudinal axis between a proximal end and a distal end. The bone plate defines a plurality of through holes extending through the elongated portion. A plurality of fasteners are configured to extend through one or more of the plurality of through holes in the bone plate and configured to secure the bone plate to the bone. The proximal end of the elongate portion has an arcuate configuration.

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Appl. No.: 18/644686

Filed: April 24, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240293161 A1	Sep. 05, 2024

Related U.S. Application Data

continuation parent-doc US 17660035 20220421 US 11986225 child-doc US 18644686
continuation parent-doc US 16795640 20200220 US 11331128 20220517 child-doc US 17660035
continuation-in-part parent-doc US 16440409 20190613 US 11278332 20220322 child-doc US

16795640
continuation-in-part parent-doc US 15871183 20180115 US 11141204 20211012 child-doc US 16440409
continuation-in-part parent-doc US 15456642 20170313 US 10420596 20190924 child-doc US 15871183
continuation-in-part parent-doc US 15238773 20160817 US 10383668 20190820 child-doc US 15456642
us-provisional-application US 62554700 20170906

Publication Classification

Int. Cl.: A61B17/17 (20060101); A61B17/16 (20060101); A61B17/80 (20060101)

U.S. Cl.:

CPC A61B17/8061 (20130101); A61B17/1615 (20130101); A61B17/1782 (20161101);

Field of Classification Search

CPC: A61B (17/1728); A61B (17/1615-1617)

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Primary Examiner: Chang; Olivia C

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 17/660,035 filed on Apr. 21, 2022, which is a continuation of U.S. patent application Ser. No. 16/795,640 filed on Feb. 20, 2020, which is a continuation-in-part of U.S. patent application Ser. No. 16/440,409, filed Jun. 13, 2019, which is a continuation-in-part of U.S. patent application Ser. No. 15/871,183, filed Jan. 15, 2018, which is a continuation-in-part of U.S. patent application Ser. No. 15/456,642, filed Mar. 13, 2017, which is a continuation-in-part of U.S. patent application Ser. No. 15/238,773, filed on Aug. 17, 2016. This

application also claims the benefit of U.S. Provisional Application Ser. No. 62/554,700, filed on Sep. 6, 2017. The contents of each of these applications are incorporated herein by reference in their entirety for all purposes.

FIELD

(1) The present disclosure relates to surgical devices and stabilization systems, for example, for trauma applications, and more particularly, for stabilization of distal radius and ulna fractures.

BACKGROUND

(2) Bone fractures are often repaired by internal fixation of the bone, such as diaphyseal bone, using one or more plates. The plate is held against the fractured bone with screws, for example, which engage the bone and heads which provide a compressive force against the plate. The plate and bone are thus forced against each other in a manner that transfers load primarily between a bone contacting surface of the plate and the bone surface to reinforce the fractured bone during healing. This manner of plating generally creates relatively low stress concentration in the bone, as there may be a large contact area between the plate and the diaphyseal bone surface permitting transfer of load to be dispersed. There may be a desire to use locking screws, non-locking screws, or a combination of both that are able to dynamically compress the bone. Of course, the designs of the plates, types of screws, and locking and/or non-locking capabilities may vary based on the location and type of fracture.

(3) The three long bones of the upper extremity are the humerus, radius, and ulna. In the case of radial fracture fixation, a volar approach may be suitable for plating certain fracture types. There remains a need, however, for improved plating systems for anatomical articular reduction and stable fixation of the radius.

SUMMARY

(4) To meet this and other needs, devices, systems, and methods of bone stabilization are provided, for example, for radius or ulna stabilization. The stabilization systems may include one or more plates and one or more fasteners. Although generally described with reference to the radius or ulna, it will be appreciated that the stabilization systems described herein may be used or adapted to be used for the fixation of other long bones as well, such as the humerus, femur, tibia, etc.

(5) According to one embodiment, a stabilization system includes a bone plate and a plurality of fasteners. The bone plate comprises an elongated portion extending along a longitudinal axis, an enlarged head portion, and a transition region connecting the elongated portion to the enlarged head portion, wherein the transition region is curved and connect to an end portion of the enlarged head portion, the bone plate comprising a plurality of through holes extending through the enlarged head portion, the transition region, and the elongated portion. The fasteners are configured to extend through one or more of the plurality of through holes in the bone plate and configured to secure the bone plate to the bone.

(6) The fasteners may include locking fasteners (e.g., configured to lock to the plate), non-locking fasteners (e.g., configured to provide dynamic compression of the bone), polyaxial fasteners (e.g., configured to be inserted at a plurality of angles or trajectories), fixed angle fasteners (e.g., configured to be inserted at a fixed angle or trajectory), or any other suitable fasteners known in the art.

(7) In some instances, the locking fasteners may include fasteners having self-forming threads on a head portion of the fasteners, which are configured to lock to at least one of the plurality of through holes on the plate.

(8) According to another embodiment, a stabilization system configured to stabilize a radius includes a bone plate, a plurality of fixed angle fasteners, a polyaxial fastener, and a fastener. The bone plate comprises an elongated portion extending along a longitudinal axis, an enlarged head portion, and a transition region connecting the elongated portion to the enlarged head portion, wherein the transition region is curved and connect to an end portion of the enlarged head portion,

the bone plate comprising a plurality of fixed angle holes positioned in general alignment along the elongated portion, a polyaxial hole positioned proximate to the end portion of the enlarged head portion connected to the transition region, and an elongated slot on the elongated portion. The fixed angle fasteners are configured to be received in the fixed angle holes, the plurality of fixed angle fasteners configured to be aimed at a radio-carpal joint and a distal radio-ulnar joint. The polyaxial fastener is configured to be received in the polyaxial hole, the polyaxial fastener configured to be aimed at a radial styloid. The fastener is configured to be received in the elongated slot, wherein the elongated slot allows for proximal-distal and medial-lateral adjustment of the plate.

(9) According to another embodiment, a stabilization system for stabilizing a bone includes a bone plate and a plurality of fasteners. The bone plate has an upper surface and a lower surface configured to contact the bone, wherein the lower surface comprises one or more recesses configured to reduce contact between the plate and a surface of the bone. The bone plate comprises an elongated portion extending along a longitudinal axis, an enlarged head portion, and a transition region connecting the elongated portion to the enlarged head portion, wherein the transition region is connect to an end portion of the enlarged head portion and the other end portion of the enlarged head portion is a free end, the bone plate comprising a plurality of through holes extending through the enlarged head portion, the transition region, and the elongated portion. The plurality of fasteners are configured to extend through one or more of the plurality of through holes in the bone plate and configured to secure the bone plate to the bone.

(10) According to another embodiment, the stabilization system may include a bone plate having an elongated portion extending along a longitudinal axis between a proximal end and a distal end. The bone plate defines a plurality of through holes extending through the elongated portion. A plurality of fasteners are configured to extend through one or more of the plurality of through holes in the bone plate and configured to secure the bone plate to the bone. The proximal end of the elongate portion has an arcuate configuration.

(11) According to another embodiment, the stabilization system may include a bone plate having an upper surface and a lower surface configured to contact the bone. The bone plate includes a plurality of through holes extending through the elongated portion. At least one of the through holes is a locking through hole which defines an upper tapered portion extending from the upper surface and a lower tapered portion extending from the lower surface with a deformation area defined between the upper tapered portion and the lower tapered portion. A plurality of fasteners are configured to extend through one or more of the plurality of through holes in the bone plate and configured to secure the bone plate to the bone. At least one of the fasteners includes self-forming threads on a head portion of the fastener which are configured to deform the deformation area and lock to one of the locking through holes on the plate.

(12) According to another embodiment, the stabilization system may include a bone plate having an upper surface and a lower surface configured to contact the bone. The bone plate further has a first portion and a second portion with the second portion extending at an angle relative to the first portion. The first portion of the bone plate includes an opening for receiving a fixation member and the second portion of the plate includes at least one hook having an arcuate configuration.

(13) According to yet another embodiment, one or more methods of installing a stabilization system may include aligning a bone plate against the volar side of the radial bone, and inserting one or more fasteners through the bone plate and into the bone to stabilize the radius and repair the fracture.

(14) Also provided are kits for the stabilization systems including bone plates of varying sizes and orientations, fasteners including locking fasteners, non-locking, compression fasteners, polyaxial fasteners, fixed angle fasteners, or any other suitable fasteners, drill guides, k-wires, and other components for installing the same.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) A more complete understanding of the present invention, and the attendant advantages and features thereof, will be more readily understood by reference to the following detailed description when considered in conjunction with the accompanying drawings wherein:
- (2) FIGS. 1A-1K depict stabilization systems according to embodiments including volar distal radius bone plates and a plurality of bone fasteners;
- (3) FIG. 2 is a top perspective view of two fasteners engaged with combination holes according to an embodiment;
- (4) FIG. 3 is a close-up view of an alternative version of a combination hole according to another embodiment;
- (5) FIGS. 4A-4C show a perspective view, top view, and cross-section view, respectively, of another embodiment of a combination hole;
- (6) FIGS. 5A-5C show a perspective view, top view, and cross-section view, respectively, of another embodiment of a combination hole;
- (7) FIGS. 6A-6C show a perspective view, top view, and cross-section view, respectively, of another embodiment of a hole for receiving a fastener;
- (8) FIGS. 7A-7C show a perspective view, top view, and cross-section view, respectively, of another embodiment of a combination hole;
- (9) FIGS. 8A-8C show a perspective view, top view, and cross-section view, respectively, of another embodiment of separate locking and non-locking holes;
- (10) FIGS. 9A-9D show a perspective view, a top view, a cross-section view, and a perspective view with a locking fastener, respectively, according to another embodiment of a plate including three overlapping locking and non-locking holes;
- (11) FIGS. 10A-10B show perspective views of a plate according to another embodiment with locking and non-locking functionality;
- (12) FIGS. 11A-11E shows alternative locking screw and openings in plates according to yet another embodiment;
- (13) FIGS. 12A and 12B depict a perspective view and cross-section view of an alternative version of a plate with blocking screws;
- (14) FIGS. 13A and 13B depict a fastener according to another embodiment with self-forming threads configured to form threads in the opening of a plate;
- (15) FIG. 13C illustrates an opening in a plate which is usable with the fastener;
- (16) FIGS. 14A and 14B depict an opening in a plate according to one embodiment having a windswept cut configured to receive the self-forming threads of the fastener of FIGS. 13A-13B;
- (17) FIGS. 15A and 15B depict an opening in a plate according to another embodiment having a knurled cut configured to receive the self-forming threads of the fastener of FIGS. 13A-13B;
- (18) FIGS. 15C and 15D depict an opening in a plate according to another embodiment having triangular cuts configured to receive the self-forming threads of the fastener of FIGS. 13A-13B;
- (19) FIGS. 16A and 16B depict an opening in a plate according to another embodiment having a polygonal cut configured to receive the self-forming threads of the fastener of FIGS. 13A-13B;
- (20) FIG. 17A depicts an alternative opening in a plate according to another embodiment;
- (21) FIG. 17B depicts another alternative opening in a plate according to yet another embodiment;
- (22) FIGS. 18A-18D depict a plate assembly according to one embodiment where a locking or non-locking fastener may be positioned at an angle or perpendicular to the plate;
- (23) FIG. 19 depicts a stabilization system according to one embodiment including a volar distal radius dia-meta bone plate;
- (24) FIG. 20 depicts a stabilization system according to one embodiment including an acute dorsal

bone plate.

(25) FIG. **21** is a cross sectional view of the dorsal bone plate of FIG. **20**;

(26) FIG. **22** depicts a stabilization system according to one embodiment including an oblique dorsal bone plate;

(27) FIG. **23** depicts dorsal bone plates of FIGS. **20** and **22** with fixation screws;

(28) FIG. **24** is a close up view of a portion of the dorsal bone plates of FIGS. **20** and **22**;

(29) FIG. **25** depicts a stabilization system according to one embodiment including a lateral bone plate;

(30) FIG. **26** is a cross sectional view of the lateral bone plate of FIG. **25**;

(31) FIG. **27** depicts the lateral bone plate of FIG. **25** with fixation screws;

(32) FIG. **28** depicts a stabilization system according to one embodiment including a bridge bone plate;

(33) FIG. **29** depicts a stabilization system according to one embodiment including a lunate facet hook plate;

(34) FIG. **30** depicts the lunate facet hook plate of FIG. **29** with a fixation screw;

(35) FIG. **31** depicts the volar distal radius plate of FIG. **1** used with the lunate facet hook plate of FIG. **30**;

(36) FIG. **32** depicts an alternative stabilization system including a lunate facet hook plate;

(37) FIG. **33** depicts the volar distal radius plate of FIG. **1** used with the lunate facet hook plate of FIG. **32**;

(38) FIGS. **34A-34E** depict an illustrative method of installing the lunate facet hook plate of FIG. **32**;

(39) FIGS. **35A-35E** depict an illustrative method of installing the lunate facet hook plate of FIG. **32** with the volar distal radius plate of FIG. **1**;

(40) FIGS. **36A-36B** depict another illustrative method of installing the lunate facet hook plate of FIG. **32** with the volar distal radius plate of FIG. **1**;

(41) FIGS. **37** and **38** depict a lunate facet hook plate reduction instrument;

(42) FIGS. **39** and **40** depict a backpack drill guide for use with at least the volar distal radius plates of FIG. **1**;

(43) FIGS. **41A** and **41B** depict a stabilization system according to one embodiment including a neck and head bone plate;

(44) FIGS. **42A** and **42B** depict an illustrative method of installing the bone plate of FIGS. **41A** and **41B**;

(45) FIG. **43** depicts a backpack drill guide in accordance with embodiments of the present disclosure for use with at least the volar distal radius plates of FIG. **1**;

(46) FIG. **44** depicts a backpack drill guide in accordance with embodiments of the present disclosure for use with at least the volar distal radius plates of FIG. **1**;

(47) FIG. **45** depicts a backpack drill guide in accordance with embodiments of the present disclosure for use with at least the volar distal radius plates of FIG. **1**;

(48) FIG. **46** depicts a cross-sectional view of the backpack drill guide of FIG. **43**;

(49) FIG. **47** depicts a side view of the backpack drill guide of FIG. **45**;

(50) FIG. **48** depicts a side view of a screw for use with a backpack drill in accordance with embodiments of the present disclosure;

(51) FIG. **49** depicts a side view of a spring drill guide in accordance with embodiments of the present disclosure;

(52) FIG. **50** depicts a side view of a targeting sleeve in accordance with embodiments of the present disclosure;

(53) FIGS. **51A** and **51B** depict side views of deployment of a backpack drill guide and spring drill guide with a volar distal radius plate in accordance with embodiments of the present disclosure;

(54) FIG. **52** depicts a cross-sectional view of a backpack drill guide in accordance with

embodiments of the present disclosure;

(55) FIG. 53A depicts a side view of the backpack drill guide of FIG. 52;

(56) FIG. 53B depicts a side view of a cam screw for use with the backpack drill of FIG. 52;

(57) FIG. 53C depicts a dowel pin for use with the cam screw of FIG. 53B and the backpack drill of FIG. 52;

(58) FIGS. 54A and 54B depict the cam screw of FIG. 53B in an unlocked position in the backpack drill guide;

(59) FIG. 54C depicts the cam screw of FIG. 53B in a locked position in the backpack drill guide;

(60) FIG. 55 depicts a top view of a cam screw disposed in a backpack drill guide in accordance with embodiments of the present disclosure; and

(61) FIG. 56 depicts a side view of a targeting sleeve in accordance with embodiments of the present disclosure.

DETAILED DESCRIPTION

(62) Embodiments of the disclosure are generally directed to devices, systems, and methods for bone stabilization, especially radius stabilization. Specifically, embodiments are directed to volar distal radius stabilization systems including a bone plate configured to sit against the volar side of the radial bone. The fasteners may be configured to secure the bone plate to the radius. Still other embodiments are directed to different types of holes and fasteners configured to provide locking and/or compression to the bone.

(63) The bone plate may be comprised of titanium, stainless steel, cobalt chrome, carbon composite, plastic or polymer-such as polyetheretherketone (PEEK), polyethylene, ultra high molecular weight polyethylene (UHMWPE), resorbable polylactic acid (PLA), polyglycolic acid (PGA), combinations or alloys of such materials or any other appropriate material that has sufficient strength to be secured to and hold bone, while also having sufficient biocompatibility to be implanted into a body. Similarly, the fasteners may be comprised of titanium, cobalt chrome, cobalt-chrome-molybdenum, stainless steel, tungsten carbide, combinations or alloys of such materials or other appropriate biocompatible materials. Although the above list of materials includes many typical materials out of which bone plates and bone fasteners are made, it should be understood that the bone plates and fasteners comprised of any appropriate material are contemplated.

(64) The embodiments of the disclosure and the various features and advantageous details thereof are explained more fully with reference to the non-limiting embodiments and examples that are described and/or illustrated in the accompanying drawings and detailed in the following description. The features of one embodiment may be employed with other embodiments as the skilled artisan would recognize, even if not explicitly stated herein. Descriptions of well-known components and processing techniques may be omitted so as to not unnecessarily obscure the embodiments of the disclosure. The examples used herein are intended merely to facilitate an understanding of ways in which the disclosure may be practiced and to further enable those of skill in the art to practice the embodiments of the disclosure. Accordingly, the examples and embodiments herein should not be construed as limiting the scope of the disclosure, which is defined solely by the appended claims and applicable law. Moreover, it is noted that like reference numerals represent similar features and structures throughout the several views of the drawings.

(65) Volar Distal Radius Plate System

(66) Referring now to the drawing, FIGS. 1A-1K depict embodiments of a volar distal radius stabilization system **100** including a bone plate **110** configured to sit against the volar side of the radial bone and one or more bone fasteners **130** configured to be received in the bone plate **110** and secured to the radius **102**. The radius **102** or radial bone is one of the two large bones of the forearm, the other being the ulna. The radius **102** extends from the lateral side of the elbow to the thumb side of the wrist and runs parallel to the ulna, which exceeds it in length and size. Near the wrist, the distal end **104** of the radius **102** is large and of quadrilateral form. Although generally

described with reference to the radius **102**, it will be appreciated that the stabilization systems described herein may be used or adapted to be used for the fixation of other long bones as well, such as the humerus, femur, tibia, etc.

(67) The bone plate **110** extends from a first end **112** configured to be positioned proximate to a shaft portion of radius **102** to a second end **114** configured to be positioned proximate to the distal end **104** of the radius **102**. The plate **110** includes a top surface **116** and an opposite, bottom surface **118** configured to contact adjacent bone. The top and bottom surfaces **116**, **118** are connected by opposite side surfaces extending from the first to second ends **112**, **114** of the plate **110**. The bottom surface **118** of the plate **110** includes an anatomic contour configured to follow the best approximation of average distal radial anatomy, flaring up slightly along the radial column and more significantly along the intermediate column of the plate **110**. The plate **110** is designed to sit low and have a generally low profile proximal portion. The thickness of the plate **110** may generally be about 2 mm along the shaft and distal intermediate column, tapering to a thickness of 2.5 mm along the distal radial column which allows for the severe angle of the radial styloid fastener. The watershed line of the volar distal radius defines the border between the radiocarpal (RC) joint and the volar surface of the radius **102**. A chamfer at the second end **114** on the distal radius column of the plate **110** may help to ensure minimal tendon disruption, for example of the flexor pollicis longus and flexor carpi radialis, by maintaining a lower profile over the tendon sites.

(68) The bone plate **110** includes an elongated portion **140** extending along a longitudinal axis L, having a length greater than its width. The elongated portion **140** is configured to contact the shaft of the radius **102**. The elongated portion **140** may terminate at the first end **112** with a taper such that it has a width and/or thickness less than the remainder of the elongated portion **140**. A transition region **144** may connect the elongated portion **140** to an enlarged head portion **142**. The transition region **144** may extend along an axis T which is generally angled relative to the axis L of the elongated portion **140**. The transition region **144** may extend at an angle X relative to the elongated portion **140**. The angle X of the transition region **144** relative to the elongated portion **140** may range from about 10-60°, about 20-50°, about 30-40°, about 40-50°, or another appropriate angle. The transition region **144** may generally form a curve from the elongated portion **140** to an end of the enlarged head portion **142**.

(69) The transition region **144** may connect to an end portion of the enlarged head portion **142** and the other end portion of the enlarged head portion **142** may be a free end. In other words, the opposite end portion of the enlarged head portion **142**, not connected to the transition region **144**, is not connected to any other portion of the plate **110**. The free end of the enlarged head portion **142** may be separated a distance from the transition region **144** and the elongated portion **140** of the plate **110**.

(70) The enlarged head portion **142** or a portion thereof is configured to contact the distal end **104** of the radius **102**. The enlarged head portion **142** has a width greater than the width of the elongated portion **140**. The enlarged head portion **142** extends along an axis A at an angle Y relative to the transition region **144**. The angle Y of the head portion **142** relative to the transition region **144** may range from about 10-60°, about 20-50°, about 30-40°, about 40-50°, or another appropriate angle. Accordingly, the axis A of the enlarged head portion **142** may be transverse to the axis L of the elongated portion **140**. In some embodiments, the axis A of the enlarged head portion **142** may be generally perpendicular to the axis L of the elongated portion **140**. As best seen in FIG. 1C, the bone plates **110** may be available in a variety of lengths and sizes based on the anatomy of the patient. The plates **110** are configured to sit against the volar side of the radial bone **102**. The plates **110** are configured in both left and right designs, in a mirrored configuration, in order to address the anatomy of both the left and right arms of the patient.

(71) As best seen in FIGS. 11 and 1J, the bottom surface **118** of the plate **110** may include a plurality of recesses **119** located along the elongated portion **140** between the fastener openings **120**. In the embodiment shown in FIG. 1I, the recesses **119** are in the form small partial bores in the

lateral surface, which are configured to facilitate bending of the plate **110**. The recesses **119** remove material such that the plate **110** shield stress from the fastener openings **120**, discouraging hole warping effect during recontouring of the plate **110**. The recesses **119** may also provide attachment points for plate placement instrumentation (not shown). In the embodiment shown in FIG. **1J**, the recesses **119** are in the form of scallop cuts having partially cylindrical valleys cut around a periphery of the bottom surface **118** of the plate **110**. This again shields stress from the fastener openings **120** during bending, discouraging hole warping effects while recontouring the plate **110**. This also reduces contact between the plate **110** and the bone surface, thereby helping to preserve blood supply to the bone and prevent osteonecrosis. In addition to or in place of the recesses **119**, a plurality of dimples, best seen in FIG. **1K**, may be positioned along the bottom surface **118** of the plate **110** (e.g., along the entire bottom surface **118** or a portion thereof) to further reduce contact between the plate **110** and bone surface, further helping to preserve blood supply and prevent osteonecrosis.

(72) The plate **110** includes one or more through openings **120** configured to receive one or more bone fasteners **130**. The openings **120** extend through the body of the plate **110** from the top surface **116** to the bottom surface **118**. The openings **120** may include cylindrical openings, conical openings, elongated openings, threaded openings, textured openings, non-threaded and/or non-textured openings, and the like. The openings **120** may allow for locking of the fastener **130** to the plate **110** or may allow for movement and dynamic compression of the bone. The plate **110** may comprise any suitable number of openings **120** in any suitable configuration. These openings **120** allow surgeons more flexibility for fastener placement, based on preference, anatomy, and fracture location. Surgeons may have differing opinions as to the number, location, and types of fasteners **130**. Further, complexity of fracture location and shape makes having as many locations for fasteners **130** as possible necessary. This design offers surgeons a versatile method to achieve higher accuracy in placement of the fasteners **130**.

(73) The openings **120** may be configured to receive one or more bone fasteners **130**. The fasteners **130** may include locking fasteners, non-locking fasteners, or any other fasteners known in the art. The fasteners **130** may comprise bone screws or the like. The fasteners **130** may also include other fasteners or anchors configured to be secured or engaged with bone, such as nails, spikes, staples, pegs, barbs, hooks, or the like. The fasteners **130** may include fixed and/or variable angle bone screws. The fastener **130** may include a head portion **132** and a shaft portion **134** configured to engage bone. For a locking fastener **130**, the shaft portion **134** may be threaded such that the fastener **130** may be threaded into the bone. The head portion **132** may include a textured area, such as threads, around its outer surface sized and configured to engage with the opening **120**, for example, and corresponding threads in the opening **120** in order to lock the fastener **130** to the plate **110**. In the alternative, for a non-locking fastener **130**, the head portion **132** may be substantially smooth to allow for dynamic compression of the bone.

(74) In one embodiment, the enlarged head portion **142** of the plate **110** includes a plurality of holes **120A** are aligned so that their nominal trajectories follow the articular surfaces of both the radio-carpal joint and the distal radio ulnar-joint. This allows the fasteners **130A** to buttress and support the articular surfaces during fracture reconstruction. As shown in the embodiment in FIGS. **1A-1C**, the plate **110** may have a single row of holes **120A** generally in alignment and a secondary hole **120A** positioned on the transition region **144**. FIG. **1C** depicts one embodiment of the plate **110** (right most plate **110**) having a first, distal row of holes **120A** generally in alignment and a second row of holes **120A** generally in alignment. The second row of holes **120A** may receive fasteners **130A** with trajectories converging with the distal row screw trajectories. In an alternative version of the stabilization system **100**, shown in FIGS. **1D** and **1E**, the elongated portion **140** directly transitions into the enlarged head portion **142** and the enlarged head portion **142** is increased in dimension in order to receive the second row of the fasteners **130A**.

(75) The holes **120A** may be fixed openings configured to accept fixed angle fasteners **130A** that

can be secured into the distal end **104** of the radius **102**. The screw holes **120A** and screw heads **132** may have mating conical threads that lock the screw **130A** in both angular and axial alignment to prevent collapse and backout. The fasteners **130A** may have predetermined trajectories based on the orientations of the openings **120A**. An upper portion of the holes **120A** may be tapered **128** to allow for the proper positioning of each of the fasteners **130A**. Each of the fasteners **130A** may be angled along a different trajectory than the other respective fasteners **130A**. Some of the fasteners **130A** may have a greater angulation than other respective fasteners **130A**.

(76) The enlarged head portion of the plate **110** further include a hole **120B** configured to receive fastener **130B** with a trajectory having the severe angle necessary to reach the tip of the radial styloid. An upper portion of the hole **120B** may be tapered **128** and a portion of the plate **110** around the hole **120B** may be enlarged or increased in thickness to allow for the proper angle of the fasteners **130B** to be achieved. The fastener **130B** may be in the form of a polyaxial bone screw, which may be generally larger (e.g., in length and/or diameter) than the other fasteners **130** securing the plate **110** to the bone. The fasteners **130A**, **130B** are optionally cannulated to allow for precise placement with a k-wire (not shown) if desired by the surgeon. In some embodiments, the fasteners **130A**, **130B** may include polyaxial screws having self-forming threads that work by displacement of the plate material, which are described in more detail herein.

(77) The plate **110** also include one or more holes **120C** present along the elongated portion **140** of the plate **110** and configured to accommodate a compression fastener **130C**. As best seen in FIG. **1G**, the holes **120C** may offer a sliding slot for proximal-distal adjustment of the plate **110** during provisional placement. The slot **120C** may allow for proximal adjustment, distal adjustment, and/or medial-lateral adjustment of the plate **110**. This allows surgeons to optimally center the plate position along the bone prior to locking screw insertion. The hole or holes **120C** may be elongated along the longitudinal axis **L** of the elongated portion **140** as well as elongated, relative to the fastener **130C**, from lateral side to lateral side. The elongated hole or holes **120C** may have varying lengths and/or widths. Preferably, the length is greater than the width of the slot **120C**. In the alternative embodiment of the slot **120C'** illustrated in FIG. **1H**, adjustment markings **125** are provided along each longitudinal side of the slot **120C'**. The adjustment markings **125** may include be applied via ink, etching, for example, via laser etching, or other suitable means. The adjustment markings **125** are spaced in increments, for example, equally spaced increments, such as 1 mm increments, to assist with accurate proximal/distal adjustment of the plate **110**.

(78) The hole **120C** may be configured to accommodate non-locking, compression screws **130C**, the heads of which have a spherical underside, so the screw **130C** may be placed at varying angles. The compression screw **130C** can be inserted and preliminarily tightened to secure the plate **110** to the bone. As the screw **130C** is inserted eccentrically in to the hole **120C**, the screw **130C** slides down the slot **120C**, displacing the plate **110** and the bone as well. The compression screw **130C** may have a shorter length and/or a smaller diameter than the screws **130A** and/or **130B**. If the plate **110** needs to be adjusted later, the screw **130C** can be loosened and the plate **110** can be shifted in the proximal, distal, and/or medial-lateral directions. This slot **120C** also accommodates reduction of the radius **102** by inserting a longer compression screw **130C** and pulling the bone to the plate **110**.

(79) The plate **110** may include one or more holes **120D** present along the elongated portion **140** of the plate **110** configured to secure the plate **110** to the shaft of the radius **102**. The holes **120D** may be configured to accommodate fixed and/or variable angle fasteners **130D**. For locking fasteners **130D**, the screw holes **120D** and screw heads **132** may have mating conical threads that lock the screw **130D** in both angular and axial alignment to prevent collapse and backout. An upper portion of the holes **120D** may be tapered **128**, for example, around the perimeter of the hole **120D**, to allow for the proper positioning of each of the fasteners **130D**. For non-locking fasteners **130D**, the head portion **132** may be substantially smooth to allow for dynamic compression of the bone.

(80) The plate **110** including head portion **142** and/or the elongated portion **140** may further

comprise a plurality of openings **124** configured to receive one or more k-wires (not shown). The k-wire holes **124** may comprise small diameter holes (e.g., having a diameter significantly smaller than the fastener openings **120**). The k-wire holes **124** may allow preliminary placement of the plate **110** against the bone and/or to aid in reduction of the fracture. The distal k-wire holes **124** on the head portion **142** may ensure a trajectory to follow the RC joint and provide direction during insertion of the distal locking screws. The proximal k-wire holes in the elongated portion **140** of the plate **110** are arranged between fastener openings **120** and may be angled relative to the surface of the plate **110** to avoid intrusion into areas where instrumentation must pass during screw insertion. (81) In the embodiment shown in FIGS. **1D** and **1E**, the plate **110** may also comprise a window **126**. The window **126** may provide visualization of the plate **110** with respect to the radius **102** in the operating environment and on imaging (e.g., fluoroscopy). The window **126** is shown as generally an asymmetrical triangle, but it is envisioned that the window **126**, if present, may be of any suitable shape, size, and dimension.

(82) The bone plate **110** may be attached to a proximal humerus to fixate one or more bone fractures or fragments and thereby promote healing of the bone. In one embodiment, the plate **110** further restores the anatomic alignment of the radius **102**. The plate **110** may be positioned against the volar side of the radial bone. One or more k-wires may be supplied through the k-wire holes **124** to assist with preliminary placement of the plate **110**. Pilot holes may be drilled through the fastener openings **120** to prepare to receive the respective fasteners **130**. The fasteners **130A**, **130B**, **130C**, **130D** may be positioned through the respective openings **120A**, **120B**, **120C**, **120D** and into the radius **102**. The fasteners **130** may be affixed to the bone in any suitable order, number, and orientation depending on the anatomy of the bone and the fracture.

(83) Alternative Hole Configurations

(84) The fixed and variable angle, locking and non-locking openings **120**, **220** (e.g., including openings **120A**, **120B**, **120C**, **120D**) and respective fasteners **130**, **230** (e.g., including **130A**, **130B**, **130C**, **130D**) described herein may be substituted with or include one or more of the following openings **20** and/or fasteners **30**, **40**. The openings **20** and/or fasteners **30**, **40** are generally described with reference to a generic plate **10**, which may include plate **110**, **210**, **310**, **410**, **510**, **610**, or any other suitable plate design.

(85) Referring now to the drawing, FIGS. **2-18** depict alternative openings **20** in plate **10**. The openings **20** extending through the plate **10** are configured to accept locking fasteners **30**, non-locking fasteners **40**, or a combination of both locking and non-locking fasteners **30**, **40** that are able to dynamically compress the bone and/or affix the plate **10** to the bone. When plating diaphyseal bone, surgeons may use a combination of both locking and non-locking fasteners **30**, **40** that are able to dynamically compress bone and to connect the bone and the plate **10**. Dynamic compression may also be desirable to create interfragmental compression while tightening the fasteners **30**, **40**.

(86) The plate **10** includes a top surface **16** and an opposite, bottom surface **18** configured to contact adjacent bone. The plate **10** includes one or more through openings **20** configured to receive one or more bone fasteners **30**, **40**. The openings **20** extend through the body of the plate **10** from the top surface **16** to the bottom surface **18**. In the embodiments depicted in FIGS. **2-3**, for example, the openings **20** may be in the form of a combination opening that has at least two overlapping holes. As shown in FIG. **2**, the combination opening **20** includes a first hole **22** overlapping a second hole **24**. One of the holes **22** may be configured to be the locking hole **22**, thereby able to receive and secure the locking fastener **30** to the plate **10**, and the other of the holes **24** may be configured to be the dynamic compression hole **24**, thereby allowing the non-locking fastener **40** to freely move in the hole **24** and apply dynamic compression. The locking hole **22** may have one or more locking features designed to engage with a locking fastener **30**, and the dynamic compression hole **24** may be elongated, for example, along the central longitudinal axis of the plate **10**. The screw holes **22**, **24** are not constrained to parallel axes. This hole geometry may be used in

bone plates **10** to utilize either fixed angle or variable angle locking screws **30** and/or polyaxial non-locking screws **40** that can achieve dynamic compression.

(87) These openings **20** allow surgeons more flexibility for fastener placement, based on preference, anatomy, and fracture location. Surgeons may have differing opinions as to whether non-locking or locking screws **30**, **40** (or some combination of the two) should be used in diaphyseal bone. Further, complexity of fracture location and shape makes having as many locations for fasteners **30**, **40** as possible necessary. This design offers surgeons a versatile method to achieve higher accuracy in placement of locking and/or non-locking screws **30**, **40**.

(88) As best seen in FIG. 2, the locking and non-locking fasteners **30**, **40** are shown. The locking and non-locking fasteners **30**, **40** may include traditional fasteners known in the art. The locking and non-locking fasteners **30**, **40** may comprise bone screws or the like. The fasteners **30**, **40** may also include other fasteners or anchors configured to be secured or engaged with bone, such as nails, spikes, staples, pegs, barbs, hooks, or the like. The fasteners **30**, **40** may include fixed and/or variable angle bone screws.

(89) The locking fastener **30** may include a head portion **32** and a shaft portion **34** configured to engage bone. The shaft portion **34** may be threaded such that the fastener **30** may be threaded into the bone. The head portion **32** of the locking fastener **30** includes a textured area **36** around its outer surface sized and configured to engage with the locking hole **22** of the combination opening **20**. The textured area **36** may include threads, ridges, bumps, dimples, serrations, or other types of textured areas. As shown, the texture area **36** preferably includes a threaded portion extending substantially from the top of the head portion **32** to the bottom of the head portion **32** proximate to the shaft portion **34**. Thus, when the textured area **36** engages the locking hole **22**, the locking fastener **30** is thereby locked to the plate **10**.

(90) The non-locking fastener **40** includes a head portion **42** and a shaft portion **44** configured to engage bone. The shaft portion **44** may be threaded such that the fastener **40** may be threaded into the bone. The head portion **42** of the non-locking fastener **40** is substantially smooth around its outer surface such that is able to slide along the elongated compression hole **24**. Thus, the non-locking fastener **30** may be coupled to the plate **10**, but not locked thereto to enable dynamic compression of the bone. It will be recognized that the head portions **32**, **42** of the fasteners **30**, **40** may include a recess configured to receive a driver or the like.

(91) The locking hole portion **22** of the combination opening **20** includes a textured portion **26**. The textured portion **26** may include threads, ridges, bumps, dimples, serrations, knurls, or other types of textured areas. The textured portion **26** may be of the same type (e.g., mating surfaces) or different from the textured area **36** of the locking fastener **30**. As shown, the textured portion **26** is serrated or knurled along an inner portion of the hole **22**. The knurled surface may include straight, angled, or crossed lines cut or rolled into the material. In the embodiment shown in FIG. 2, the textured portion **26** extends along substantially the entire inner surface of the hole **22**. With reference to the embodiment shown in FIG. 3, the combination hole **20** is substantially the same as that shown in FIG. 2 except that the textured portion **26** the locking hole **22** now includes a thin centralized textured ribbon of material. For example, the textured portion **26** takes up about half or less of the surface area of the hole **22**. In this instance, only a portion of the textured area **36** of the head portion **32** of the locking fastener **30** engages with and locks to the textured portion **26** of the hole **22**.

(92) An upper portion of the hole **22** may be tapered **28**, without texturing, for example, to facilitate alignment of the fastener **30** with the opening **20**. As shown in FIG. 3, this tapered portion **28** is enlarged in area relative to the embodiment in FIG. 2. The hole **22** may be configured to receive a fixed or variable angle fastener **30**. The hole **22** may be generally conical in shape such that it is wider near the top surface **16** of the plate **10** and narrower toward the bottom surface **18** of the plate **10**. The tapered portion **28** and/or the textured area **26** may be conical in shape. In this embodiment, the locking hole **22** is a textured fixed angle conical hole configured to receive

locking fastener **30**. The textured holes **22** may deform as the fastener head **32** interferes with the textured portion **26** of the hole **22**, thereby providing a positive lock between the fastener **30** and the plate **10**.

(93) The second hole portion **24** of the combination opening **20** may be an elongated dynamic compression hole. The dynamic compression hole **24** may be elongated such that it has a length greater than its width. The hole **24** may be elongated along the longitudinal axis of the plate **10**. In the alternative, the hole **24** may be generally cylindrical such that the hole **24** only permits polyaxial movement of the fastener **40**. The inner surface of the hole **24** may be substantially smooth such that the non-locking fastener **40** is able to freely pivot and/or slide along the hole **24**. This provides for at least two directions of compressive force (e.g., along the longitudinal axis and perpendicular to the longitudinal axis of the plate **10**). The head portion **42** of the non-locking fastener **40** may be substantially smooth around its outer surface. The head portion **42** is sized and configured to engage with and be retained within the hole portion **24** of the combination opening **20**. The hole **24** may be configured to receive a fixed or variable angle fastener **40**. In one embodiment, the hole **24** may be generally conical in shape and/or tapered such that it is wider near the top surface **16** of the plate **10** and narrower toward the bottom surface **18** of the plate **10**. In this embodiment, the hole **24** is a smooth variable angle conical hole configured to receive the non-locking fastener **40**. The hole **24** may receive the fastener head **42** allowing movement of the fastener **40**, for example, in a polyaxial fashion and/or along the length of the hole **22**, thereby providing dynamic compression of the bone.

(94) Turning now to FIGS. **4-7**, alternative types of openings **20A-20G**, which provide for locking and/or non-locking, dynamic compression are provided. As many of the features of these openings are similar to the combination openings **20** described already for FIGS. **2-3**, only the different features will be further explained.

(95) With reference to FIGS. **4A-4C**, the combination opening **20A** is similar to combination opening **20** except that the dynamic compression hole **24A** has the same general diameter as the locking hole **22A**, and the locking hole **22A** includes a different type of textured portion **26A**. In this embodiment, the locking hole **22A** has a first diameter **D1**, and the dynamic compression hole **24A** has a second diameter **D2**. Unlike the elongated hole **24** described earlier, dynamic compression hole **24A** has substantially same diameter as the locking hole **22A**. Thus, the first and second diameters **D1**, **D2** are substantially the same. The hole **24A** may be formed by milling or drilling a sphere out of the plate **10** in the center of the circle with tapers or ramps on either side. The hole **24A** is not elongated, but is generally circular and the non-locking fastener **40** will be allowed to translate in the hole **24A** because the diameter of the head portion **42** and/or shaft (e.g., bone thread) will be smaller than the size of the hole **24A** in the plate **10**. With respect to hole **22A**, the textured portion **26A** of the hole **22A** may be in the form of a tapered thread. This tapered thread may generally correspond to a similar tapered thread on the locking fastener **30**. This hole **22A** also does not include a tapered portion, and the textured portion **26A** begins at the intersection with the top surface **16** of the plate **10**. This alternative opening **20A** also provides for the use of both locking and non-locking fasteners **30**, **40** that are able to dynamically compress bone and/or lock the plate **10** to the bone.

(96) Turning now to FIGS. **5A-5C**, the combination opening **20B** is similar to other combination openings except that the locking hole **22B** includes a different type of textured portion **26B**. The textured portion **26B** includes a series of alternating recesses and protrusions around a central portion of the hole **22B**. The recesses may be in form of a wave of alternating cutouts extending around the inner perimeter of the hole **22B**. The textured portion **26B** may lock the fastener **30** with a friction fit or may be modified during insertion of the fastener **30** to form a lock in situ. In this embodiment, the locking hole may allow for polyaxial locking. The plate **10** and the locking fastener **30** may be made of dissimilar materials having dissimilar hardness values. For example, the fastener **30** may have a higher hardness (e.g., on the Rockwell scale) relative to the plate **10**,

which may be formed of a material having a lower relative hardness value. Due to the increased hardness, the head portion **32** of the locking fastener **30** may create a thread in the plate **10** as the fastener **30** is inserted (e.g., threaded) into the hole **22B**, thereby locking the fastener **30** to the plate **10**.

(97) With reference to FIGS. **6A-6C**, the opening **20C** includes locking hole **22C** and dynamic compression hole **24C** with a more open configuration. The locking portion **22C** has a textured portion **26C** in the form of a tapered thread. This tapered thread may generally correspond to a similar tapered thread on the locking fastener **30**. The opposite portion **24C** of the opening **20C** is oblong with a ramp **25C** milled into the top surface **16** of the plate **10** to allow for dynamic compression. As best seen in FIG. **6C**, the ramp may be partially spherical in shape and extend from the top surface **16** of the plate **10** and connect to the textured portion **26C**. When viewed from above in FIG. **6B**, the ramp **25C** creates a square-like, key-hole, and/or non-hole geometry that sweeps into the tapered threaded locking hole **22C**. This alternative opening **20C** also provides for the use of both locking and non-locking fasteners **30**, **40** that are able to dynamically compress bone and/or lock the plate **10** to the bone.

(98) Turning now to FIGS. **7A-7C**, the opening **20D** includes locking hole **22D** and dynamic compression hole **24D**. These holes **22D**, **24D** are connected and close together but are not overlapping. The holes **22D**, **24D** are separated by a small portion or sliver of plate material proximate to the lower portion of the holes **22D**, **24D** (e.g., at bottom surface **18** of the plate **10** and partially extending between the holes **22D**, **24D**). The locking portion **22D** has a textured portion **26D** in the form of a tapered thread. The textured portion **26D** extends around almost the entire circumference of the hole **22D** except where connected to hole **24D**. The dynamic compression hole **24D** is elongated and has ramped portions **25D** on opposite sides of the hole **24D** to receive fastener **40**. This configuration allows for a very close population of holes **22D**, **24D** on the plate **10** while giving structural stability at the holes **22D**, **24D**.

(99) With reference to FIGS. **8A-8C**, locking hole **22E** and dynamic compression hole **24E** are adjacent, but separate from one another. The holes **22E**, **24E** are completely separated from one another by a wall **56** of plate material. The locking portion **22E** has a textured portion **26E** in the form of a tapered thread extends around the entire perimeter of the hole **22E**. The dynamic compression hole **24E** is elongated and has ramped portions **25E** on opposite sides of the hole **24E**. This configuration also allows for a very close population of holes **22E**, **24E** on the plate **10** while giving options for both locking and/or dynamic compression.

(100) Turning now to FIGS. **9A-9D**, an alternative version of opening **20F** is provided. In this embodiment, the hole construct **20F** is comprised of at least three overlapping conical threaded holes in the plate **10**. The opening **20F** includes a first, locking hole **22F**, a second hole **24F**, and a third hole **23F** arranged along a longitudinal axis of the plate **10**. The third hole **23F** is the mirror image of hole **24F** across the first locking hole **22F**. The conically threaded holes **22F**, **23F**, **24F** may or may not have parallel axes. Each hole **22F**, **23F**, **24F** may include a textured portion **26F**, for example, in the form of one or more threaded portions. Thus, the locking fastener **30** may lock to any of the holes **22F**, **23F**, **24F**. Although each of the holes **22F**, **23F**, **24F** are shown in with the textured portion **26F**, it will be appreciated that one or more of the holes **22F**, **23F**, **24F** may have a substantially smooth inner portion instead of the textured portion **26F**. The upper part of the hole construct at the first and second ends of the hole **20F** each have a ramped feature **25F** (e.g., adjacent to holes **23F** and **24F**) to allow for dynamic compression of the plate **10**. In addition, the ramped feature **25F** may span the three or more conical holes **22F**, **23F**, **24F** (e.g., around the entire perimeter of the opening **20F**).

(101) The non-locking compression fasteners **40** may have a major bone thread diameter such that the fastener **40** can translate between overlapping holes **22F**, **24F**, **23F** without interference. As best seen in FIG. **9D**, the locking fastener **30** may include a textured area **36**, for example, in the form of a thread, configured to engage with the textured portion **26F** of any of the holes **22F**, **23F**, **24F**. The

hole geometry of opening **20F** can be applied to bone plates **10** to utilize either fixed angle and/or variable angle locking screws **30** and/or polyaxial non-locking screws **40** that can achieve dynamic compression. This allows surgeons more flexibility for screw placement, based on preference, anatomy, and fracture location.

(102) Turning now to FIGS. **10A-10B**, another embodiment of opening **20G** is provided. This opening **20G** may be comprised of one elongate hole or slot extending from the top surface **16** to the bottom surface **18** of the plate **10**. A locking portion **22G** of the opening **20G** may include a textured portion **26G** having straight machine threads. The threads may extend more than 180 degrees to retain the locking fastener **30**. A non-locking portion **24G** of the opening **20G** may be positioned opposite the locking portion **22G** to complete the opening **20G**. The upper part of the opening **20G** may have one or more ramped features **25G** to allow for dynamic compression of the plate **10**. The ramp **25G** may span along the entire upper perimeter of the elongated slot **20G** or a portion thereof. The compression screws **40** may have a major bone thread diameter such that the screws **40** are able to translate along the opening **20G** without interference.

(103) With reference to FIGS. **11A-11E**, alternative embodiments of the locking fastener **30** may be used with any plate **10**. The head portion **32** of the fastener **30** may include a textured area **36** in the form of a thread, for example, to lock the fastener **30** to the plate **10**. The fastener **30** and/or plate **10** may also include one or more mechanisms to prevent back out of the fastener **30** from the plate **10**. In FIG. **11A**, the head portion **32** includes a threaded portion **36A** (e.g., having straight threads) that interface with the plate **10** and the top of the head extends larger than the threads. The head portion **32** bottoms out when the fastener **30** is fully inserted and creates preload in the fastener **30**, thus locking the fastener **30** rotationally. In FIG. **11B**, the head portion **32** includes threaded portion **36B**. The head portion **32** has a constant major diameter while the minor diameter is tapered. The thread depth may go to zero at the top of the head portion **32** of the screw **30**. The first few turns smoothly insert, but as the tapered portion of the male thread engages with the plate **10**, interference occurs, jamming and/or locking the screw **30** and preventing backout. In FIG. **11C**, a screw thread **36C** on the head portion **32**, similar to the design in FIG. **11B**, except the minor diameter of the screw **30** stays constant while the major diameter of the head portion **32** gets larger toward the top of the screw **30**. A similar jamming and locking mechanism results through tightening of the screw **30** in the plate **10**. In FIG. **11D**, the threaded portion **36D** has areas of varying pitch. In particular, a straight screw thread on the head portion **32** of the screw **30** has a similar pitch to that of the plate **10** at the bottom of the head portion **32** of the screw **30**. The pitch then increases or decreases towards the top of the head portion **32**, which thereby results in jamming of the threads and preventing unwanted backout of the screw **30**. In an alternative variation of the concept of FIG. **11D**, shown in FIG. **11E**, the opening in the plate **10** is provided with areas of varying pitch while the pitch of the threaded portion **36D** remains constant. For example, the head portion **32** may include a straight thread with a constant pitch. The upper surface of the plate **10** may include a thread pitch is similar to that of the screw **10**, but towards the bottom surface of the plate **10**, the thread pitch would either increase or decrease to lock the screw **30** to the plate **10**.

(104) Turning now to FIGS. **12A** and **12B**, the plate **10** includes an additional anti-backout feature. In this embodiment, the plate **10** includes cylindrical holes or openings **20H** configured to accept either the compression fastener **40** or the locking fastener **30**. Each opening **20H** may include a ramped portion **25H** extending around a portion or the entire perimeter of the opening **20H** to allow for dynamic compression with a compression fastener **40**. Each opening **20H** may include a cylindrical feature to provide angular stability with a locking fastener **30**. The opening **20H** may also include an angular taper **28** to cause compressive tightening between the locking fastener **30** and the cylindrical opening **20H**. Each opening **20H** has an accompanying blocking screw **46** that can be actuated to block the fastener **30**, **40** from backing out. The blocking screw **46** may extend from a first end at the top surface **16** to a second end at the bottom surface **18** of the plate **10**. The

first end of the blocking screw **46** may include a recess sized to receive an instrument to rotate the blocking screw **46** from an unblocked position to a blocked position. The blocked position may include a portion of the blocking screw **46** covering a portion of the head portion **42** of the fastener **40**, thereby further preventing backout of the fastener **40** from the plate **10**.

(105) According to yet another embodiment, the plate **10** may include one or more openings **20** configured to receive the locking fastener **30** having self-forming threads that work by displacement of the plate material to lock the fastener **30** to the plate **10**. Turning now to FIGS. **13-18**, the locking fastener **30** and alternative embodiments of the openings **20** in the plate **10** are shown. In these embodiments, the locking mechanism of the fastener **30** (e.g., bone screw) to the internal fixation plate **10** may allow for variable angle screw insertion. The fastener **30** may be inserted within an angular cone where the force required to dislodge the head portion **32** of the fastener **30** is substantially equivalent to the force required when the fastener **30** is inserted perpendicular to the plate **10**. The holes or openings **20** in the plate **10** may be shaped such that the fastener **30** may be inserted at different angles. The geometry of the opening **20** is conducive to catching the threads on the head portion **32** of the fastener **30** and to reduce the axial force necessary to initiate the thread formation.

(106) The locking mechanism includes a fastener **30** having a head portion **32** with self-forming threads that displace the plate material. The plate **10** may be made of a material softer than the fastener **30** to facilitate displacement. For example, the plate **10** may be comprised of titanium, alloys, polymers, or other materials having a lower material hardness (e.g., Rockwell hardness). The fastener **30** may be made of a harder relative material, for example, comprised of cobalt chrome, tungsten, alloys, or other materials having a higher material hardness. Preferably, the fastener **30** is comprised of a material having a strong, stiff, and high surface hardness which facilitates the thread forming process. The forming mechanism works by displacement of material rather than removal of the material of the plate **10**, thereby minimizing fragments or chips which are created from tapping.

(107) In FIGS. **13A-13B**, the locking fastener **30** includes a head portion **32** and a shaft portion **34** configured to engage bone. Although not shown, the shaft portion **34** may be threaded such that the fastener **30** may be threaded into the bone. The head portion **32** may be tapered (e.g., at an angle of about 20°) such that the fit within the opening **20** in the plate **10** becomes tighter as the fastener **30** is advanced in to the bone. The head portion **32** of the locking fastener **30** includes a textured area **36** around its outer surface sized and configured to engage an opening **20** in the plate **10**. The textured area **36** may include threads, ridges, bumps, dimples, serrations, or other types of textured areas. As shown, the textured area **36** preferably includes a threaded portion extending substantially from the top of the head portion **32** to the bottom of the head portion **32** proximate to the shaft portion **34**. The threads **36** may run generally perpendicular to the conical surface of the head portion **32**. The threaded portion **36** is in the form of self-forming threads configured to displace the plate material and create threads in the opening **20** of the plate **10**. The threaded portion has an exaggerated sharp thread peak to facilitate cutting or forming of the plate material.

(108) Turning now to FIGS. **13C-17**, alternative versions of the openings **20** are shown before being tapped with the fastener **30**. Once the fastener **30** is inserted, these openings **20** are modified based on the self-forming threads. The geometry of the openings **20** are conducive to catching the threads **36** and designed to reduce the axial force necessary to initiate the thread formation. An upper portion of the hole **20** may be tapered **28**, for example, with a conical straight tapered surface cut through the top surface **16** of the plate **10** for clearance of the head portion **32** of the fastener **30** during off angle insertion. A lower portion of hole **20** may further be tapered **29**, for example, with a conical straight tapered surface cut through the bottom surface **18** of the plate **10** for clearance of the shaft portion **34** during off angle insertion. The upper tapered portion **28** may be larger, for example, with a larger degree of taper than the lower tapered portion **29**. For example, the upper tapered portion **28** may have a taper in a range from about 60-90°, 70-80°, or 72-78°, preferably

about 70°, 75°, or 80° whereas the lower tapered portion **29** may have a taper in a range from about 50-70°, 55-65°, or 57-63°, preferably about 55°, 60°, or 65°. The upper and/or lowered tapered portions **28**, **29** may be substantially conical (e.g., FIGS. **14B**, **15B**, **15C**, **16B**) or may be segmented with more than one section, such as two separate conical sections having different diameters or degrees of taper (e.g., FIGS. **17A** and **17B**).

(109) At the intersection between the upper tapered portion **28** and the lower tapered portion **29**, a narrowed central portion, as indicated by the area **31** within the dashed lines of FIG. **13C** defines the area where thread forming takes place. The area **31** provides a concentric ring of material for material displacement and thread forming. The area **31** may have the untextured surface illustrated in FIG. **13C** or may have a textured portion **26**. As described herein, the textured portion **26** may include threads, ridges, bumps, dimples, serrations, or other types of textured areas. In the embodiment shown in FIGS. **14A-14B**, the textured portion **26** includes a windswept cut design comprised of a plurality of shallow cuts where each cut overlaps the next. For example, the windswept design may include a plurality of threadlike helical cut sweeps. Each cut has a smooth transition into the inner diameter of the hole **20** (e.g., into the upper and lower tapered portions **28**, **29**). The windswept cuts provide a positive surface for the self-forming threads to cut into, thereby helping to prevent peeling of the newly formed threads into the plate **10**.

(110) In FIGS. **15A-15D**, the textured portion **26** includes a knurled cut design. In the embodiment illustrated in FIGS. **15A-15B**, a rounded transition between the upper tapered portion **28** and the lower tapered portion **29** (e.g., the two conical cuts) provides a workable surface for the knurling process as well as a surface for the head portion **32** to be able to roll over during off-axis locking. The knurled design may include a plurality of shallow knurled grooves set in a diamond pattern (e.g., about 45°) where each cut overlaps the next. The knurled grooves allow for the self-forming threads to cut more deeply into the material and reduce the necessary axial force to begin the thread forming process. In the embodiment illustrated in FIGS. **15C-15D**, the area **31** has a textured portion **26** defined by a plurality of 360° circular swept cuts. Additionally, a series of 60° triangular cuts is made at a 17° trajectory from a plane normal to hole axis, with the same number of cuts being applied in both a clockwise and counter-clockwise fashion, creating a pattern on the inside ring of material. The cuts create “plateaus” of material protruding into the hole, as shown. While specific angles are described, the disclosure is not limited to the specific angles. The resultant geometry from the cuts provides positive surfaces to cut into, dramatically reducing the axial force necessary to lock the screw to the plate. As such, the mechanism does not rely on bone purchase to engage the threads in the head of the screw. Secondly, the material removed by the cuts allow the head threads to cut deeper by reducing the amount of material which must be formed, and reducing friction between the screw **30** and plate **10** during the forming process.

(111) FIGS. **16A-16B** depict a polygon form cut design. In this design, there is no textured portion at the transition between the upper tapered portion **28** and the lower tapered portion **29**. Instead, the narrowed central region has an overall polygonal form such that the hole **20** is neither cylindrical nor conical. The polygonal shape includes a number of sides with distinct linear section of material and rounded corners around which the form cut is allowed to sweep. For example, the polygonal shape may be substantially hexagonal (6-sided), heptagonal (7-sided), octagonal (8-sided), etc. The hole **20** may also be represented without lobe cuts, as a single concentric ring with the same geometry.

(112) In FIG. **17A**, the upper tapered portion **28** includes a conical straight tapered surface cut for clearance of the head portion **32** of the fastener **30** during off angle insertion. The upper tapered portion **28** is segmented to have an upper area with a larger area relative to a lower area proximate the transition to the lower tapered portion **29** having a narrower diameter. The central area between the upper and lower tapered portions **28**, **29**, where the thread forming process occurs, includes two peaks or concentric rings of material (e.g., a superficial ring **60** and a deep ring **62**) with a groove **27** being located in between for material removal and thread forming relief. The groove **27**

between the rings **60**, **62** may be angled, for example, in the range of about 40-80°, about 50-70°, or about 60°. The superficial ring **60** is of a slightly smaller inner diameter than the deep ring **62**, as the superficial ring **60** is responsible for supporting a majority of the cantilever loads. The deep ring **62** provides additional fixation and support during off-angle insertion as well as additional support during nominal trajectory insertion. The lower tapered portion **29** includes a straight tapered surface that provides clearance for the shaft **34** of the fastener **30** when inserted off angle. (113) The embodiment of the opening **20** in FIG. **17B** is similar to FIG. **17A**, but further includes textured portion **26** in the form of a plurality of helical swept cuts at the transition between the upper tapered portion **28** and the lower tapered portion **29**. The shallow helical cuts or windswept cuts may include a series of cuts at a steep pitch. The windswept cuts may be angled, for example, at about 50-70°, or about 60°. The same number of cuts may be made in both a clockwise and counter-clockwise fashion. The cuts may create plateaus of material protruding into the opening **20**. The resultant geometry provides positive surfaces for the fastener **30** to cut into, which can dramatically reduce the axial force necessary to lock the fastener **30** to the plate **10**. Thus mechanism does not need to rely on bone purchase in order to engage the threads in the head portion **32** of the fastener **30**. The material removed during insertion of the fastener **30** allows the self-forming threads to cut deeper by removing material which much be formed and reducing friction between the fastener **30** and the plate **10** during the forming process.

(114) FIGS. **18A-18D** depict a screw-plate assembly. The assembly, in FIG. **18C**, shows the locking fastener **30** placed at an angle, other than perpendicular, to the upper surface **16** of the plate **10**. In FIG. **18D**, a non-locking fastener **40** is placed generally perpendicular to the plate **10**. It will be appreciated that the locking fastener **30** and non-locking fastener **40** may be oriented at any appropriate angle relative to the plate **10**. The section view in FIG. **18C** shows the thread engagement with the plate **10** in which material of the plate **10** is displaced around the threads of the fastener **30**. By using the self-forming threads, the fastener **30** is able to be inserted into the plate **10** at variable angles and engages with the plate **10** with one-step locking requiring no additional steps to lock the fastener **30** to the plate **10**. The section view in FIG. **18D** show the compressive, non-locking screw **40** received in the opening **20**, without threadedly locking thereto. The non-locking screw **40** may provide for dynamic compression of the bone. Accordingly, the fasteners and openings described herein provide a wide variety of options for the surgeon, thereby providing appropriate locking and/or unlocking capability for dynamic compression depending on the desired treatment of the fracture and the bone.

(115) Dia-Meta Volar Distal Radius Plate System

(116) FIG. **19** depicts embodiments of a dia-meta volar distal radius stabilization system **200** including a bone plate **210** configured to sit against the volar side of the radial bone and one or more bone fasteners are configured to be received in the bone plate **210** and secured to the radius and radial shaft of a bone. Although generally described with reference to the radius and radial shaft of a bone, it will be appreciated that the stabilization system **200** described herein may be used or adapted to be used for the fixation of other long bones as well, such as the humerus, femur, tibia, etc.

(117) The bone plate **210** extends from a first end **212** configured to be positioned on a shaft portion of radial bone to a second end **214** configured to be positioned proximate to the distal end of the radius. The plate **210** includes a top surface **216** and an opposite, bottom surface **218** configured to contact adjacent bone. The top and bottom surfaces **216**, **218** are connected by opposite side surfaces extending from the first to second ends **212**, **214** of the plate **210**. The bottom surface **218** of the plate **210** includes an anatomic contour configured to follow the best approximation of average distal radial anatomy, flaring up slightly along the radial column and more significantly along the intermediate column of the plate **210**. The plate **210** is designed to sit low and have a generally low profile proximal portion. The thickness of the plate **210** may generally be about 2 mm along the shaft and distal intermediate column, tapering to a thickness of

2.5 mm along the distal radial column which allows for the severe angle of the radial styloid fastener. The thickness of the plate **210** may generally increase towards the first end **212** when compared to the second end **214**. In addition, the width of the plate **210** proximate the first end **212** and along the elongate portion **240** may be thicker than the width of the plate at the second end **214**. The design of plate **210** allows for an easy transition from the second end **214** of the plate **210** to the elongate portion **240** to the first end **212** of the plate **210** to address fractures proximal to the second end of the plate **214** while also providing adequate support in the radial shaft of the bone.

(118) The second end **214** of the bone plate **210** toward the elongate portion **240** of the bone plate **210** is very similar to the bone plate **110**, thus the features and disclosures set forth above relating to the bone plate **110** are equally applicable to bone plate **210** and are incorporated in their entirety herein.

(119) Looking at the elongate portion or dia-meta portion **240** of the plate **210**, the plate **210** includes one or more through openings **220** configured to receive one or more bone fasteners. The openings **220** extend through the body of the plate **210** from the top surface **216** to the bottom surface **218**. The openings **220** may include cylindrical openings, conical openings, elongated openings, threaded openings, textured openings, non-threaded and/or non-textured openings, and the like. The openings **220** may allow for locking of the fastener to the plate **210** or may allow for movement and dynamic compression of the bone. The plate **210** may comprise any suitable number of openings **220** in any suitable configuration. These openings **220** allow surgeons more flexibility for fastener placement, based on preference, anatomy, and fracture location. Surgeons may have differing opinions as to the number, location, and types of fasteners. Further, complexity of fracture location and shape makes having as many locations for fasteners as possible necessary. This design offers surgeons a versatile method to achieve higher accuracy in placement of the fasteners.

(120) The openings **220** may be configured to receive one or more bone fasteners. The fasteners may include locking fasteners, non-locking fasteners, or any other fasteners known in the art. The fasteners may comprise bone screws or the like. The fasteners may also include other fasteners or anchors configured to be secured or engaged with bone, such as nails, spikes, staples, pegs, barbs, hooks, or the like. The fasteners may include fixed and/or variable angle bone screws. The fastener may include a head portion and a shaft portion configured to engage bone. For a locking fastener, the shaft portion may be threaded such that the fastener may be threaded into the bone. The head portion may include a textured area, such as threads, around its outer surface sized and configured to engage with the opening **220**, for example, and corresponding threads in the opening **220** in order to lock the fastener to the plate **210**. In the alternative, for a non-locking fastener, the head portion may be substantially smooth to allow for dynamic compression of the bone.

(121) The plate **210** may further comprise a plurality of openings **224** configured to receive one or more k-wires (not shown). The k-wire holes **224** may comprise small diameter holes (e.g., having a diameter significantly smaller than the fastener openings **220**). The k-wire holes **224** may allow preliminary placement of the plate **210** against the bone and/or to aid in reduction of the fracture. The distal k-wire holes **224** on the head portion **242** may ensure a trajectory to follow the RC joint and provide direction during insertion of the distal locking screws. The proximal k-wire holes in the elongated portion **240** of the plate **210** are arranged between fastener openings **220** and may be angled relative to the surface of the plate **210** to avoid intrusion into areas where instrumentation must pass during screw insertion.

(122) Dorsal Plate System

(123) FIGS. **20-24** depict embodiments of a dorsal stabilization system **300** including bone plates **310**, **410** which are configured to sit against the dorsal portion of bone. One or more bone fasteners **320C** are configured to be received in the bone plates **310**, **410** to secure the plates **310**, **410** to the dorsal portion of a bone. Although generally described with reference to the dorsal portion of bone, it will be appreciated that the stabilization system **300** described herein may be used or adapted to be used for the fixation of other bones as well, such as other portions of the identified bones. It

should be noted that the same reference numerals are being used for plates **310**, **410** because the plates are similar except for their respective first ends **312** which show different opening **320** configurations. FIG. **20** shows an acute configuration and FIG. **22** shows an oblique configuration. (124) As shown in FIGS. **20-22**, the plates **310**, **410** each have a body that extends from a first end **312** to a second end **314**. The plates **310**, **410** each include a top surface **316** and an opposite, bottom surface **318** configured to contact adjacent bone. The top and bottom surfaces **316**, **318** are connected by opposite side surfaces extending from the first to second ends **312**, **314** of the plate **310**. Although the plate **310**, **410** are shown having a generally longitudinal body, it will be appreciated that any suitable shape and contouring of the plates may be provided depending on the location and type of fracture to be plated.

(125) The bone plates **310**, **410** include one or more openings **320**. The openings **320** extend through the plate **310**, **410** from the upper surface **316** to the bottom surface **318** and are configured to accept locking fasteners and non-locking fasteners **320C**. When using the plates **310**, **410** with bone, surgeons may use only locking, only non-locking or a combination of both locking and non-locking fasteners to connect the bone and the plates **310**, **410**. The openings **320** may be in the form of any of the openings discussed above with respect to the volar distal radial plate system, the diaphyseal plate system, and the alternative hole configurations.

(126) The plates **310**, **410** also include one or more slots **320C** present along the elongated portion **340** of the plates **310**, **410** and configured to accommodate a sliding fastener **322C**, shown in FIGS. **23** and **24**. As best seen in FIGS. **20-24**, the slot **320C** may offer a sliding slot for proximal-distal adjustment of the plates **310**, **410** during provisional placement. The slot **320C** may allow for proximal adjustment, distal adjustment, and/or medial-lateral adjustment of the plates **310**, **410**. This allows surgeons to optimally center the plate position along the bone prior to locking screw insertion. The slot **320C** may be elongated along a longitudinal axis of the elongated portion **340** as well as elongated, perpendicular to the longitudinal axis, from lateral side to lateral side. The elongated slot **320C** may have varying lengths and/or widths. Preferably, the length is greater than the width of the slot **320C**. The plates **310**, **410** may include etch lines adjacent to slot **320C** for more accurate adjustment of the plate **310** when being positioned on bone.

(127) As best seen in FIGS. **20** and **22**, plates **310**, **410** also may include a plurality of side relief cuts or scalloped edging **322** along the length of the plates **310**, **410** which allows the plates **310**, **410** to be bent, for example, in three dimensions. The side relief cuts or scalloped edges **322** may be in the form of one or more curves having a widened portion along the sides of the plates **310**, **410** and a narrowed portion towards the center of the plates **310**, **410**. The side relief cuts or scalloped edges **322** may be positioned between consecutive openings **320**. The plurality of relief cuts or scalloped edges **322** may form a scalloped or wavy profile along the side edges of the plates **310**, **410**. As a result, the plates **310**, **410** are able to be shaped to a multi-contour surface without warping the openings **320**.

(128) The plates **310**, **410** may further comprise a plurality of openings **324** configured to receive one or more k-wires (not shown). The k-wire holes **324** may comprise small diameter holes (e.g., having a diameter significantly smaller than the fastener openings **320**). The k-wire holes **324** may allow preliminary placement of the plates **310**, **410** against the bone and/or to aid in reduction of the fracture.

(129) Lateral Plate

(130) FIGS. **25-27** depict embodiments of a lateral stabilization system **500** including bone plate **510** which is configured to sit against the lateral portion of bone to address fractures on the side of the radius. One or more bone fasteners **520C** are configured to be received in the bone plate **510** to secure the plate **510** to the lateral portion of a radius of a bone. Although generally described with reference to the lateral portion of the radius of the bone, it will be appreciated that the stabilization system **500** described herein may be used or adapted to be used for the fixation of other bones, such as long bones, as well as other portions of the identified bones.

(131) The plate **510** has a body that extends from a first end **512** to a second end **514**. The plate **510** includes a top surface **516** and an opposite, bottom surface **518** configured to contact adjacent bone. The top and bottom surfaces **516**, **518** are connected by opposite side surfaces extending from the first to second ends **512**, **514** of the plate **510**. Although the plate **510** is shown having a generally longitudinal body, that contours or radius upwardly to accommodate distal radius bony anatomy, it will be appreciated that any suitable shape and contouring of the plates may be provided depending on the location and type of fracture to be plated.

(132) The bone plate **510** includes one or more openings **520**. The openings **520** extend through the plate **510** from the upper surface **516** to the bottom surface **518** and are configured to accept locking fasteners and non-locking fasteners **520C**. When using the plate **510** with bone, surgeons may use only locking, only non-locking or a combination of both locking and non-locking fasteners to connect the bone and the plate **510**. The openings **520** may be in the form of any of the openings discussed above with respect to the volar distal radial plate system, the dia-meta plate system, the dorsal plates and the alternative hole configurations.

(133) The plate **510** also includes one or more slots **520C** present along the elongated portion **540** of the plate **510** and configured to accommodate a sliding fastener **522C**, shown in FIG. 27. As best seen in FIGS. 25-26, the slot **520C** may offer a sliding slot for proximal-distal adjustment of the plate **510** during provisional placement. The slot **520C** may allow for proximal adjustment, distal adjustment, and/or medial-lateral adjustment of the plate **510**. This allows surgeons to optimally center the plate position along the bone prior to locking screw insertion. The slot **520C** may be elongated along a longitudinal axis of the elongated portion **540** as well as elongated, perpendicular to the longitudinal axis, from lateral side to lateral side. The elongated slot **520C** may have varying lengths and/or widths. Preferably, the length is greater than the width of the slot **520C**. The plate **510** may include etch lines adjacent to slot **520C** for more accurate adjustment of the plate **510** when being positioned on bone.

(134) As best seen in FIGS. 25 and 27, plate **510** also may include a plurality of side relief cuts or scalloped edging **522** along a portion of the length of the plate **510** which allows that portion of the plate **510** to be bent, for example, in three dimensions. The side relief cuts or scalloped edges **522** may be in the form of one or more curves having a widened portion along the sides of the plate **510** and a narrowed portion towards the center of the plate **510**. The side relief cuts or scalloped edges **522** may be positioned between consecutive openings **520**. The plurality of relief cuts or scalloped edges **522** may form a scalloped or wavy profile along the side edges of the plate **510**. As a result, a portion of the plate **510** is able to be shaped to a multi-contour surface without warping the openings **520**.

(135) The plate **510** may further comprise a plurality of openings **524** configured to receive one or more k-wires (not shown). The k-wire holes **524** may comprise small diameter holes (e.g., having a diameter significantly smaller than the fastener openings **520**). The k-wire holes **524** may allow preliminary placement of the plate **519** against the bone and/or to aid in reduction of the fracture.

(136) Bridge Plate

(137) FIG. 28 depicts an embodiment of a stabilization system **600** including bone plate **610** which acts as an internal fixator for high energy comminuted distal radius fractures. The plate **610** is placed dorsally and extends from the third or second metacarpal to approximately a third to half way down the radius. One or more bone fasteners are configured to be received in the bone plate **610** to secure the plate **610** to the desired portions of bone. Although generally described with reference to the radius and metacarpals, it will be appreciated that the stabilization system **600** described herein may be used or adapted to be used for the fixation of other bones, such as long bones, as well as other portions of the identified bones.

(138) The plate **610** has a body that extends from a first end **612** to a second end **614**. The plate **610** includes a top surface **616** and an opposite, bottom surface **618** configured to contact adjacent bone. The top and bottom surfaces **616**, **618** are connected by opposite side surfaces extending

from the first to second ends **612**, **614** of the plate **610**. Although the plate **610** is shown having a generally longitudinal body that is generally planar, it will be appreciated that any suitable shape and contouring of the plates may be provided depending on the location and type of fracture to be plated.

(139) The bone plate **610** includes one or more openings **620**. The openings **620**, which are located proximate the first end **612** and the second end **614**, extend through the plate **610** from the upper surface **616** to the bottom surface **618** and are configured to accept locking fasteners and non-locking fasteners. When using the plate **610** with bone, surgeons may use only locking, only non-locking or a combination of both locking and non-locking fasteners to connect the bone and the plate **610**. The openings **620** may be in the form of any of the openings discussed above with respect to the volar distal radial plate system, the dia-meta plate system, the dorsal plates, the lateral plates and the alternative hole configurations.

(140) Lunate Facet Hook Plate

(141) FIGS. **29-36** depict embodiments of a stabilization system **700** including hook plate **710**, **710'** which is designed for fracture patterns that involve the volar ulnar corner of the distal radius. The plate **710**, **710'** may be used as a stand-alone stabilization plate, as shown in FIGS. **32** and **34** or may be used in combination with a volar distal radius plate **110**, as shown in FIGS. **31**, **33** and **35-36**.

(142) When the plate **710** is used alone, the hooks **712** of the plate are embedded or tapped into bone to prevent the shifting of the plate in a lateral or medial direction. It is contemplated that there may one, two, or more hooks **712**. The plate **710** also includes an opening **720** to receive a fixation screw **714**, which may aid in further fixation of the plate **710** the bone and the fracture site.

(143) When the plate **710** is used with the volar distal radius plate, the plate **710** is configured and dimensioned such that it can be slidably placed under a pre-positioned volar distal radius plate **110**. The opening **720** will align with an opening **120** on the volar distal radius plate **110** such that a fastener will pass through the opening **120** on the volar distal radius plate **110** and the opening **720** on the plate **710**. The opening **720** can accept a locking screw or a non-locking screw.

(144) Referring to FIGS. **32** and **34A-34E**, the plate **710'** is similar to the previous embodiment and includes a pair of hooks **712'** extending from the body **716** of the plate **710'** with a transition area **713** therebetween. The hooks **712'** have an arcuate configuration such that the hooks **712'** complement the curvature of the rim of the lunate facet (see FIGS. **32** and **34E**). The transition area **713** narrows between the elongate body **716** and the hooks **712'**. An elongate slot **720** extends through the elongate body **716** and is configured to receive a fixation screw **714**. As seen in FIG. **34E**, the elongate body **716** may have an initial curvature or be bent to complement the contour of the bone **102**.

(145) Referring to FIGS. **34A-34E**, an illustrative method of installing the plate **710'** as a stand-alone stabilization plate will be described. As shown in FIG. **34A**, an inserter **730** is utilized to hold plate **710'** and apply the plate **710'** to the bone **102** with the hooks **712'** positioned over the lunate facet. A tamp **732** is struck with a mallet **734** or the like to tamp the hooks **712'** in or over the facet as shown in FIG. **34B**. With the plate **710'** in position, a hole is drilled through the slot **720**, for example, utilizing a drill **738** with a bit passing through a soft tissue protector **736** as illustrated in FIG. **34C**. It may be desired to position the hole in a proximal portion of the slot **720**. After the hole is drilled, the desired screw length may be determined utilizing a depth gauge or the like (not shown). The desired fixation screw **714** may then be inserted into the hole using a driver **740** or the like, as shown in FIG. **34D**. Non-locking screws lag the plate **710'** to the bone **102** and help to maintain fracture compression and avoiding last translation of the plate **710'** distally. FIGS. **32** and **34E** illustrate the final construct of the stabilization plate. Fluoroscopy, as shown in FIG. **34E**, may be utilized to confirm proper screw **714** placement.

(146) Turning to FIGS. **35A-35E**, a first illustrative method of installing the plate **710'** along with a volar distal radius plate **110** will be described. In this illustrative method, the plate **710'** is

positioned prior to the volar plate insertion. Initially, the plate **710'** is positioned relative to the bone **102** and the hooks **712'** are tamped into position in a manner similar to that described with respect to FIGS. **34A** and **34B**. With the plate **710'** in position, a k-wire **742** is inserted through the slot **720** and into the bone **102** as shown in FIG. **35A**. The volar plate **110** is then positioned such that the k-wire **742** extends through one of the screw holes **120A** in the head portion **142** of the volar plate **110** as shown in FIG. **35B**. The inserter **730** may be utilized to hold and manipulate the volar plate **110**. The volar plate **110** is slid along the k-wire **742** such that the volar plate **110** is properly positioned on the bone **102** and is covering the hook plate **710'** as shown in FIG. **35C**. When positioned correctly, the ulnar-most subchondral locking screw hole **120A** aligns with the slot **720** of the hook plate **710'**. With the volar plate **110** positioned properly, a fixation screw **714** is inserted through the screw hole **120A** and the slot **720** and secured within the bone. FIGS. **33** and **35E** illustrate the final construct of the stabilization assembly. Fluoroscopy, as shown in FIG. **35E**, may be utilized to confirm proper screw **714** placement and proper placement of the hook plate **710'**.

(147) Referring to FIGS. **36A** and **36B**, another illustrative method of installing the plate **710'** along with a volar distal radius plate **110** will be described. In this illustrative method, the volar plate **110** has already been installed and thereafter it is determined that a hook plate **710'** is desired, for example, when an unstable lunate facet fracture is identified after the volar plate fixation. Initially, at least one of the fixation screws securing the head portion of the volar plate **110** is removed (not shown). Thereafter, the elongate body **716** of the hook plate **710'** is slid behind the volar plate **110** and the hooks **712'** are tamped into position as shown in FIG. **36A**. With the plate **710'** in position, a k-wire **742** is inserted through the slot **720** and into the bone **102** as shown in FIG. **35A**. The volar plate **110** is then positioned such that the k-wire **742** extends through one of the screw holes **120A** in the head portion **142** of the volar plate **110** as shown in FIG. **35B**. Once the hook plate **710'** positioned properly, a fixation screw **714** is inserted through the screw hole **120A** and the slot **720** and secured within the bone as illustrated in FIG. **36B**. The final construct of the stabilization assembly will be as shown in FIGS. **33** and **35E**. Again, fluoroscopy, as shown in FIG. **35E**, may be utilized to confirm proper screw **714** placement and proper placement of the hook plate **710'**.

(148) FIGS. **37** and **28** show a lunate facet hook plate reduction instrument **810**. The instrument is capable of being connected to any quick connect handle known in the industry, such as the AO quick-connect handle. The reduction instrument **810** utilizes a two-piece contact surface that is capable of capturing a lunate facet hook plate and releasing the hook plate when it is positioned in the desired location and orientation.

(149) FIGS. **39** and **40** depict a drill guide **910** that can be attached to second end **114** of the volar distal radius plate **110**. The drill guide **910** may include a plurality of cannulated openings **912** which correspond to each of the respective openings **120** in the plate **110**. The drill guide **910** openings **912** may be configured in order to drill the pilot holes at the appropriate trajectories for each opening **120**, and subsequently receive the respective fasteners at the correct trajectories. The drill guide **910** may also include a plurality of k-wire openings **914** which match with the k-wire openings in the plate **110**. The drill guide **910** may be secured to the plate **110** with one or more fasteners or may be secured to the plate **110** through an integrated connection system such as a thumb screw, an interference fit, etc. The fastener may thread into the plate **110** or otherwise temporarily secure the drill guide **910** to the plate **110**. The drill guide **910** may be pre-assembled to the plate **110** or may be attached at any other suitable time before or during the surgery. The fastener may be secured, for example, in the operating room, via thumb or hexalobular fastener, to attach the drill guide **910** to the plate **110**. After the pilot holes are drilled, the drill guide **910** may then be removed and the fasteners positioned through the respective openings **120**. The drill guide **910** may be relatively slim in thickness, for example, not protruding more than 10 mm above the plate **110**, to prevent impinging on soft tissue.

(150) Neck and Head Plate

(151) FIGS. **41A** and **41B** depict an embodiment of a lateral stabilization system **1000** including

bone plate **1010** which is configured to treat fractures of the ulnar neck and head. One or more bone fasteners **1030** are configured to be received in the bone plate **1010** to secure the plate **1010** to the neck and head portion of the ulna **102**. The plate **1010** has a body that extends from a first end **1012** to a second end **1014**. The plate **1010** includes a top surface **1016** and an opposite, bottom surface **1018** configured to contact adjacent bone. The top and bottom surfaces **1016**, **1018** are connected by opposite side surfaces extending from the first to second ends **1012**, **1014** of the plate **1010**. The first end **1012** of the plate **1010** preferably has a chamfer which allows distal placement of the plate **1010**. The second end **1014** of the plate **1010** has an arcuate configuration which complements the curvature of the distal portion of the ulna **102**. The plate **1010** has a generally longitudinal body, that contours or radius upwardly slightly to accommodate distal radius bony anatomy, however, it will be appreciated that any suitable shape and contouring of the plates may be provided depending on the location and type of fracture to be plated.

(152) The bone plate **1010** includes one or more openings **1020**. The openings **1020** extend through the plate **1010** from the upper surface **1016** to the bottom surface **1018** and are configured to accept locking fasteners and non-locking fasteners **1030**. When using the plate **1010** with bone, surgeons may use only locking, only non-locking or a combination of both locking and non-locking fasteners to connect the bone and the plate **1010**. The openings **1020** may be in the form of any of the openings discussed above with respect to the volar distal radial plate system, the dia-meta plate system, the dorsal plates and the alternative hole configurations. The proximal most opening **1020A** preferably is a polyaxial screw hole which is angled distally. Such a configuration helps to prevent screw impingement on the articular surface.

(153) The plate **1010** also includes one or more slots **1020C** present along the elongated portion **1040** of the plate **1010** and configured to accommodate a sliding fastener **1030C**. The slot **1020C** has a configuration similar to the slot **120C'** illustrated in FIG. 1H and may offer a sliding slot for proximal-distal adjustment of the plate **1010** during provisional placement. The slot **1020C** may allow for proximal adjustment, distal adjustment, and/or medial-lateral adjustment of the plate **1010**. This allows surgeons to optimally center the plate position along the bone prior to locking screw insertion. The slot **1020C** may be elongated along a longitudinal axis of the elongated portion **1040** as well as elongated, perpendicular to the longitudinal axis, from lateral side to lateral side. The elongated slot **1020C** may have varying lengths and/or widths. Preferably, the length is greater than the width of the slot **1020C**. The plate **1010** may include etch lines adjacent to slot **1020C** for more accurate adjustment of the plate **1010** when being positioned on bone.

(154) As best seen in FIG. 41A, plate **1010** also may include a plurality of side relief cuts or scalloped edging **1022** along a portion of the length of the plate **1010** which allows that portion of the plate **1010** to be bent, for example, in three dimensions. The side relief cuts or scalloped edges **1022** may be in the form of one or more curves having a widened portion along the sides of the plate **1010** and a narrowed portion towards the center of the plate **1010**. The side relief cuts or scalloped edges **1022** may be positioned between consecutive openings **1020**. The plurality of relief cuts or scalloped edges **1022** may form a scalloped or wavy profile along the side edges of the plate **1010**. As a result, a portion of the plate **1010** is able to be shaped to a multi-contour surface without warping the openings **1020**.

(155) The plate **1010** may further comprise a plurality of openings **1024** configured to receive one or more k-wires (not shown). The k-wire holes **1024** may comprise small diameter holes (e.g., having a diameter significantly smaller than the fastener openings **1020**). The k-wire holes **1024** may allow preliminary placement of the plate **1010** against the bone and/or to aid in reduction of the fracture. Alternatively, as illustrated in FIG. 42A, a drill guide **744** may be utilized to direct a drill bit of a drill **738** through the openings **1020** through the plate **1010**. For example, the proximal most opening **1020A** and the distal most opening **1020D** may be utilized for positioning of k-wires **742** to provisionally hold the plate **1010** in place. The placement of the k-wires **742** may be confirmed utilizing fluoroscopy as illustrate in FIG. 42B.

(156) According to one embodiment, the ulna plate **1010** may be used for fixation of an unstable ulna following distal radius repair. Using a subcutaneous ulnar approach, the patient's arm may be positioned on a hand table with the elbow flexed. The forearm may be positioned to expose the subcutaneous border of the ulna. A longitudinal incision may be created distally and proximally. The interval between the extensor carpi ulnaris (ECU) and the flexor carpi ulnaris (FCU) may be split to expose the ulnar shaft. The plate **1010** may be applied dorsally if desired. The fracture may be reduced and the reduction may be confirmed, for example, with fluoroscopy. The speed locking drill guide **744** may be used to place k-wires **742** in the distal and proximal screw holes to provisionally hold the plate in position. A hole may be drilled through the center of the positioning slot, and a screw may be positioned therein, thereby allowing for adjustment of the plate **1010** proximal-distal and/or medial-lateral for optimal placement. The remaining screws may be predrilled and placed and the k-wires may be replaced with locking screws.

(157) Targeting Guide and Spring Drill Guide

(158) FIG. **43** depicts a targeting guide **4300** in accordance with embodiments of the present disclosure. The targeting guide **4300** includes a body **4302** having an upper portion **4304** configured to be attached to the second end **114** of the volar distal radius plate **110** (e.g., the volar distal plate **110** depicts in FIGS. **1D** and **1E**) and a lower portion **4306** having a collet (shown in FIGS. **46** and **47**) configured to couple the targeting guide **4300** to the plate **110**, and a neck portion **4305** extending between the upper and lower portions **4304**, **4306**. The targeting guide **4300** may include a plurality of cannulated openings **4312** which correspond to each of the respective openings **120** in the plate **110**. The openings **4312** may be configured in order to drill the pilot holes at desired trajectories for each opening **120**, and subsequently receive the respective fasteners at the desired trajectories. The targeting guide **4300** may also include a plurality of k-wire openings **4314** which match with the k-wire openings **124** in the plate **110**.

(159) FIG. **44** depicts an alternative targeting guide **4400** in accordance with embodiments of the present disclosure. The targeting guide **4400** is substantially similar to the targeting guide **4300** except that the targeting guide **4400** has a lower profile design than the targeting guide **4300**. This lower profile design is achieved by eliminating the neck portion such that a lower portion **4406** having the collet is adjacent an upper portion **4404** of the body **4402**. FIG. **45** depicts another embodiment of a targeting guide **4500** according to the present disclosure. The targeting guide **4500** is also substantially similar to the targeting guides **4300** and **4400** except that the targeting guide **4500** has an even lower profile than the targeting guide **4400** to advantageously minimize disruption of soft tissue, maximize visualization of the surgical field, and enable insertion and removal of the targeting guide without removal of any k-wires extending through the distal radius plate **110**. To facilitate such functionality, a body **4502** of the targeting guide **4500** having an upper portion **5404** and a lower portion **4506** is configured so that it does not surround the k-wire openings **124** of the volar distal radius plate **110**.

(160) FIGS. **46-48** illustrate how the targeting guide (i.e., the targeting guides **4300**, **4400**, **4500**) is coupled to the volar distal radius plate **110** in accordance with embodiments of the present disclosure. FIG. **46** is a cross-sectional view of the targeting guide **4300** coupled to the distal radius plate **110**. FIG. **47** is a side view of the targeting guide **4500**. Although specific targeting guides are illustrated in FIGS. **46** and **47**, the following description of the coupling of the targeting guide applies to the other embodiments of the targeting guide discussed herein. As shown more clearly in FIGS. **46** and **47**, each targeting guide (**4300**, **4400**, and **4500**) includes a pair of protrusions **4602** (shown in FIG. **46**) and **4702** (shown in FIG. **47**) that extend towards the distal radius plate to serve as standoffs which provide improved rigidity when the targeting guide is coupled to the distal radius plate. Also shown in FIGS. **46** and **47** is a collet **4604** which is configured to be inserted in one of the openings **120** to couple the targeting guide to the distal radius plate **110**.

(161) The collet **4604** includes a plurality of arms **4606**. In some embodiments, the plurality of arms **4606** may include 3 to 8 arms. The arms **4606** are configured to be deflected radially inward

by the walls of the opening **120** during insertion of the collet **4604** into the opening **120** and subsequently return to their initial undeflected position after the collet **4604** is pushed beyond a portion **4608** of the opening **120** having a reduced diameter. In some embodiments, the collet **4604** includes a reduced diameter portion **4610** configured to receive and mate with the portion **4608** of the opening **120** and a flange **4612** disposed distal to the reduced diameter portion **4610** and extending radially outward from the collet **4604**. The reduced diameter portion **4610** and the flange **4612** are configured to lock the collet **4604** (and the targeting guide) to the distal radius plate **110**. The collet **4604** advantageously provides a rigid method of coupling the targeting guide to the distal radius plate without the need for threaded holes.

(162) In some embodiments, the targeting guide may further include a screw **4800** configured to be inserted into a central opening **4614** of the collet **4604** to prevent inadvertent deflection of the collet **4604** radially inward and thus, removal of the targeting guide from the distal radius plate **110**. In some embodiments, a distal end of the screw **4800** may be capped to prevent binding of the screw in the central opening **4614**. As shown more clearly in FIGS. **43-45**, the screw **4800** includes an engagement feature **4802** (e.g., a hexalobular shaped feature) formed in the head **4804** of the screw **4800** to facilitate the engagement of a driver (not shown) with the screw **4800**. The targeting guides **4300**, **4400**, **4500** advantageously allow for attachment of the guide to the distal radius plate on a back table or intra-operatively with the plate already fixed to the bone.

(163) FIG. **49** depicts a spring drill-guide **4900** in accordance with embodiments of the present disclosure. The spring drill-guide **4900** includes a drill bit **4901** extending from a proximal end **4902** to a distal end **4903**. The drill bit **4901** includes an attachment portion **4905** at the proximal end **4902** that facilitates attachment of the drill bit **4901** to a drill (not shown). The attachment portion **4905** has a larger diameter than the remainder of the drill bit **4901**. The spring drill-guide **4900** further includes a compression element such as, for example, a helical spring **4904** and a sleeve **4906** coupled to a distal end of the helical spring **4904**. The spring **4904** extends from a distal end of the attachment portion to the distal end **4903** of the drill bit **4901**. The spring **4904** has a diameter smaller than a diameter of the attachment portion such that an axial force applied to the spring **4904** compresses the spring. The drill bit **4901** extends through the helical spring **4904** and the sleeve **4906** such that in an initial state where the spring is not compressed, the distal end **4903** extends through the sleeve **4906**. The sleeve **4906** is configured such that when the spring drill-guide **4900** is inserted into one of the openings (e.g., **4310**) of the targeting guide (e.g., **4300**), the sleeve **4906** remains in the opening as the drill bit **4901** is inserted further through the opening, thus resulting in the compression of the spring **4904**. FIG. **51A** depicts the initial insertion of the spring drill-guide **4900** into one of the openings of the targeting guide. FIG. **51B** depicts the further insertion of the drill bit **4901** through the opening while the sleeve **4906** remains in the opening and the spring **4904** is compressed. Because the sleeve **4906** is part of the spring drill-guide **4900**, the number of steps and hands needed to place screws using a targeting guide is advantageously reduced. Furthermore, the spring drill-guide **4900** also advantageously modulates the speed at which the drill bit **4901** is plunged into the bone (due to the spring forces), thus reducing the likelihood of a surgeon accidentally plunging the drill bit too far. FIG. **50** illustrates a targeting sleeve **5000** which may alternatively be used with a conventional drill bit (not shown) and any one of the targeting guides **4300**, **4400**, **4500**.

(164) FIG. **52** depicts a cross-sectional view of a targeting guide **5200** coupled to the volar distal radius plate **110** in accordance with embodiments of the present disclosure. FIG. **53A** is a side view of the targeting guide **5200**. Similar to the targeting guide **4300** discussed above, the targeting guide **5200** includes a pair of protrusions **5202** (one shown in FIG. **52**) that extend towards the distal radius plate to serve as standoffs which provide improved rigidity when the targeting guide is coupled to the distal radius plate. Also shown in FIGS. **52** and **53A** is a retaining arm **5204** disposed adjacent to a counterbored hole **5205** and extending downward from the targeting guide **5200** and configured to be inserted in one of the openings **120** to couple the targeting guide to the

distal radius plate **110**. Similar to the targeting guides discussed above, the targeting guide **5200** may include a plurality of cannulated openings **5213** which correspond to each of the respective openings **120** in the plate **110**. The openings **5213** may be configured in order to drill the pilot holes at desired trajectories for each opening **120**, and subsequently receive the respective fasteners at the desired trajectories. The targeting guide **5200** may also include a plurality of k-wire openings **5214** which match with the k-wire openings **124** in the plate **110**.

(165) Although only one retaining arm is shown, the targeting guide **5200** may alternatively include a plurality of arms. In some embodiments, the retaining arm **5204** is configured to be deflected radially inward by the walls of the opening **120** during insertion of the retaining arm **5204** into the opening **120** and subsequently return to its initial undeflected position after the retaining arm **5204** is pushed beyond a portion **5208** of the opening **120** having a reduced diameter. In some embodiments, the retaining arm **5204** includes a detent **5210** configured to receive and mate with the portion **5208** of the opening **120** and a flange **5212** disposed distal to the detent **5210** and extending radially outward from the retaining arm **5204**.

(166) FIG. **53B** depicts a side view of a cam screw **5300** for use with the targeting guide **5200** in accordance with embodiments of the present disclosure. FIG. **53C** depicts a dowel pin **5500** for use with the targeting guide **5200** in accordance with embodiments of the present disclosure. To lock the targeting guide **5200** to the plate **110**, a cam screw **5300** is inserted into the counterbored hole **5205** of the targeting guide **5200**. In some embodiments, the cam screw **5300** includes a head **5302** and a shaft **5304** extending from the head **5302**. Similar to the retaining arm **5204**, the shaft **5304** includes a radially outwardly extending flange **5306** extending from a distal end **5308** of the shaft **5304**. As shown in FIG. **53B**, the head **5302** includes a cam surface **5310** that extends radially from the head **5302** in a direction opposite to that of the flange **5306**. During initial placement of the targeting guide **5200** onto the plate **110**, the cam screw **5300** is disposed in the counterbored hole **5205** such that the flange **5306** is adjacent the flange **5212** of the retaining arm **5204**, as shown in FIG. **54A**. In some embodiments, the flange **5212** includes a detent **5406** configured to receive the flange **5306** of the cam screw **5300**, as shown in FIG. **54B**. As a result, the retaining arm **5204** is easily inserted into the opening **120** of the plate. To lock the targeting guide **5200** to the plate **110**, the cam screw **5300** is rotated so that the flange **5306** is disposed on an opposite side of the opening **120** beneath the portion **4608** of the opening **120** having a reduced diameter, as shown in FIG. **54C**. The extension of the flange **5306** of the cam screw **5300** and the flange **5212** of the retaining arm **5204** in substantially opposite directions beneath the portion **4608** of the opening **120** lock the targeting guide **5200** to the plate **110** in a similar manner to the collet **4604** described above.

(167) As shown in FIG. **55**, in some embodiments, the cam screw **5300** may be retained in the counterbored hole **5205** by a dowel pin **5502**. The dowel pin **5502** extends through a hole **5312** (shown in FIG. **53A**) that extends transversely to the counterbored hole **5205**. To prevent the cam screw **5300** from backing out, the dowel pin **5502** extends above the cam screw **5300**. The cam screw **5300** also includes an engagement feature **5315** configured to be engaged by a driver to rotate the cam screw **5300**.

(168) Once the targeting guide **5200** is locked onto the plate **110**, holes may be drilled into a bone to which the plate **110** is to be coupled using, for example, the spring drill-guide **4900**. FIG. **56** illustrates a targeting sleeve **5600** which may alternatively be used with a conventional drill bit (not shown) and any one of the targeting guides **4300**, **4400**, **4500**, **5200**. In some embodiments, the targeting sleeve **5600** includes a handle portion **5602** configured to be grasped by a user's hand and a guide portion **5604** configured to allow a drill bit to pass therethrough.

(169) Although the invention has been described in detail and with reference to specific embodiments, it will be apparent to one skilled in the art that various changes and modifications can be made without departing from the spirit and scope of the invention. Thus, it is intended that the invention covers the modifications and variations of this invention provided they come within the scope of the appended claims and their equivalents. It is expressly intended, for example, that

all ranges broadly recited in this document include within their scope all narrower ranges which fall within the broader ranges. It is also intended that the components of the various devices disclosed above may be combined or modified in any suitable configuration.

Claims

1. A method for positioning a volar distal plate on bone, comprising the steps of: providing a targeting guide having: a body having an upper portion configured to be disposed on a head portion of the volar distal radius plate and a lower portion configured to couple the targeting guide to the volar distal radius plate; a counterbored hole extending through the body; a retaining arm disposed adjacent the counterbored hole and extending from the lower portion of the body and into an opening in the volar distal radius plate, wherein the retaining arm includes a detent configured to receive and mate with a portion of the opening of the volar distal radius plate and a flange extending radially outward from a distal end of the retaining arm; and a cam screw disposed through the counterbored hole, wherein the cam screw includes a head and a shaft extending from the head, wherein the shaft includes a radially outwardly extending flange extending from a distal end of the shaft; drilling pilot holes and inserting fasteners at desired trajectories through a plurality of cannulated openings and through corresponding respective openings of the volar distal radius plate; securing the volar distal radius plate to bone.
2. The method of claim 1, further comprising: a plurality of k-wire openings corresponding to a plurality of k-wire openings in the volar distal radius plate.
3. The method of claim 1, wherein in the unlocked position, the radially outwardly extending flange of the cam screw is disposed adjacent the flange of the retaining arm, and wherein in the locked position, the radially outwardly extending flange of the cam screw extends in a substantially opposite direction as the flange of the retaining arm.
4. The method of claim 1, further comprising: a dowel pin extending through an opening formed in the targeting guide, wherein the opening extends transverse to the counterbored hole, and wherein the dowel pin extends above the cam screw to prevent the cam screw from backing out.
5. The method of claim 1, wherein the body further includes a neck portion extending between the upper portion and the lower portion.
6. The method of claim 1, wherein the upper portion is adjacent the lower portion.
7. The method of claim 1, wherein the body does not surround k-wire holes in the volar distal radius plate.
8. A method for stabilizing a bone, the method comprising: providing a system comprising: a bone plate, the bone plate comprising an elongated portion extending along a longitudinal axis, an enlarged head portion, a transition region connecting the elongated portion to the enlarged head portion, and a plurality of through holes extending through the elongated portion and the head portion; a targeting guide configured to be coupled to the bone plate, comprising: a body having an upper portion configured to be disposed on the enlarged head portion of the bone plate and a lower portion configured to couple the targeting guide to the bone plate; a plurality of cannulated openings corresponding to respective through holes of the bone plate, wherein the cannulated openings are configured to facilitate drilling pilot holes and inserting fasteners at desired trajectories through the through holes of the bone plate; a counterbored hole extending through the body; a retaining arm disposed adjacent the counterbored hole and extending from the lower portion of the body and into an opening in the bone plate, wherein the retaining arm includes a detent configured to receive and mate with a portion of the opening of the bone plate and a flange extending radially outward from a distal end of the retaining arm; and a cam screw disposed through the counterbored hole, wherein the cam screw includes a head and a shaft extending from the head, wherein the shaft includes a radially outwardly extending flange extending from a distal end of the shaft, a spring drill-guide configured to be coupled to a drill and comprising: a drill bit

extending from a proximal end to a distal end, a compression element; and a sleeve coupled to a distal end of the compression element, wherein the drill bit extends through the compression element and the sleeve such that, in an initial state where the compression element is not compressed, only the distal end of the drill bit extends through the sleeve, using the system to stabilize the bone.

9. The method of claim 8, wherein the compression element is a helical spring.

10. The method of claim 8, wherein the targeting guide further comprises: a plurality of k-wire openings corresponding to a plurality of k-wire openings in the bone plate.

11. The method of claim 8, wherein in the unlocked position, the radially outwardly extending flange of the cam screw is disposed adjacent the flange of the retaining arm, and wherein in the locked position, the radially outwardly extending flange of the cam screw extends in a substantially opposite direction as the flange of the retaining arm.

12. The method of claim 8, wherein the targeting guide further comprises: a dowel pin extending through an opening formed in the targeting guide, wherein the opening extends transverse to the counterbored hole, and wherein the dowel pin extends above the cam screw to prevent the cam screw from backing out.

13. The method of claim 8, wherein the body of the targeting guide further includes a neck portion extending between the upper portion and the lower portion.

14. The method of claim 8, wherein the upper portion of the targeting guide is adjacent the lower portion of the targeting guide.

15. The method of claim 14, wherein the body of the targeting guide does not surround k-wire holes in the bone plate.
